M. Hesham

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SUMMARY

AI Engineer with 5+ years of Python backend experience building production microservices, LLM pipelines, and agentic applications. Proven track record shipping RAG systems, multi-tool AI agents, and event-driven automation at scale. MS in Computer Science with hands-on experience across cloud infrastructure, distributed systems, applied AI, and modern AWS-based architectures.

SKILLS

Languages: Python, SQL, C++, JavaScript
Frameworks & Tools: FastAPI, LangChain, Docker, PostgreSQL, ChromaDB
Cloud & Infrastructure: AWS (EC2, S3, Lambda, SQS), Linux
AI / ML: RAG Pipelines, LLM Serving (vLLM, Ollama), Vector Embeddings, Agentic Systems, Semantic Search

EXPERIENCE

Texas State University, San Marcos, TX — Jan 2024 – Dec 2025
Graduate Assistant

- Created a Python and Flask internal portal serving 70+ department users that let students quickly schedule tutoring sessions, send inquiries online, and access detailed real-time schedules, reducing staff response time to student requests by 30%.
- Developed a Python pipeline automating semester reports for a 15-person department, removing 10+ hours of manual work.

J&M Jewelry, San Marcos, TX — Jan 2025 – Apr 2025
Software Engineer Intern

- Improved order accuracy by 25% for 500+ monthly customers by engineering an LLM-powered extraction pipeline in Python that parsed natural language chat into the validated new ring design parameters, enforcing deterministic output schemas.
- Reduced invalid configuration outputs by 30% by implementing 25+ validation rules in Python enforcing inventory constraints.

Orange Business, Cairo, Egypt — Dec 2019 – Dec 2023
Software Engineer

- Engineered a real-time Python ingestion pipeline that aggregated distributed performance metrics across 50+ enterprise environments, cutting mean time to detection by 35% and improving alert fidelity and incident response speed.
- Reduced undetected network outages by 40% by engineering automated health-monitoring scripts in Python that continuously polled 200+ routers and switches across enterprise networks, triggering real-time alerts on anomaly detection.
- Eliminated manual compliance review for 3 EMEA enterprise clients by architecting FastAPI microservices that automated end-to-end data validation workflows, cutting processing time by 70% and enabling same-day audit readiness.

PROJECTS

RAG Document Assistant | Python, FastAPI, LLaMA 3.1, LangChain, ChromaDB, Ollama — **GitHub**

- Enabled natural language querying over private documents by building an end-to-end RAG pipeline using FastAPI, LLaMA 3.1, LangChain, and ChromaDB, delivering source-cited answers across 7indexed chunks with efficient retrieval accuracy.
- Eliminated cloud API dependencies by serving LLaMA 3.1 8B locally via Ollama with 4-bit quantization, enabling fully offline inference on consumer hardware at zero cost per query while maintaining practical deployment flexibility and portability.
- Reduced hallucinated outputs by constraining generation to semantically retrieved context using 384-dimensional vector embeddings, grounding all responses in source documents with stronger answer reliability and contextual precision.

Weather Agent App | Python, LangChain, OpenAI, Tool Calling, FastAPI — **GitHub**

- Built a production-ready multi-tool AI agent using LangChain and OpenAI function that autonomously routes natural language weather queries to specialized API tools, achieving reliable real-time responses across orchestration.
- Designed a modular tool-calling architecture with typed input/output schemas and error-handling fallbacks, enabling reliable agent behavior across diverse query types, edge cases, and dynamic real-time tool execution and recovery workflows.
- Deployed via FastAPI supporting multi-turn conversations with persistent session context and structured tool execution logs.

Distributed XGBoost Training & Performance Profiling | Python, Dask, XGBoost, SLURM, OpenMP — **GitHub**

- Engineered distributed ML training pipelines using Python, Dask, and XGBoost on TACC's Lonestar6 HPC cluster, processing the NYC Taxi dataset across multi-node GPU environments with scalable orchestration and reproducible execution workflows.
- Profiled parallel scaling efficiency across worker configurations using SLURM job scheduling and OpenMP thread metrics, identifying XGBoost's internal multi-threading as the bottleneck through systematic performance analysis and tuning.

EDUCATION

Texas State University, San Marcos, TX — Dec 2025
Master of Science, Computer Science | Unity Scholarship Winner | GPA: 3.92

Arab Academy for Science, Technology, Cairo, Egypt — Jun 2019
Bachelor of Engineering, Electronics and Communications